Fiscal Year:	FY 2023	**Task Last Updated:**	FY 02/12/2026
PI Name:	Blakely, Eleanor A Ph.D.		
Project Title:	Simulation of GCR-Induced Harderian Gland and Lung Tumorigenesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Berkeley	**State:**	CA
Zip Code:	94720	**Congressional District:**	13
Comments:			
Project Type:		**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	02/04/2016	**End Date:**	03/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2022 and grant number is now 80JSC02IT0017, per PI (Ed., 6/7/21) NOTE: End date changed to 2/4/2022 per PI and Space Radiation element (Ed., 3/17/21) NOTE: End date is 2/4/2021 per PI (Ed., 11/7/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mao, Jian-Hua Ph.D. (Lawrence Berkeley National Laboratory) Chang, Polly Ph.D. (SRI International)		
Grant/Contract No.:	80JSC02IT0017 ; NNJ16HP22I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Increased cancer risk remains one of the primary concerns for travel into deep space and may preclude manned missions to Mars due to the large uncertainties that currently exist in estimating cancer risk from the spectrum of radiation types found in space with available human epidemiological data on radiation-induced cancer. Existing data on human risk of cancer from exposures to X- and gamma-rays must be scaled to the many types and fluxes of radiations found in space using radiation quality factors and dose-rate modification factors. Linearity of dose-response is often assumed since the shape of the dose-response at doses below 100 mSv is unknown. However, preliminary analysis of our recently completed experiments with silicon and titanium ion suggest non-linear structure in the low-dose region for both Harderian and lung tumorigenesis. The goal of this proposal is to reduce uncertainties in the estimation of particle radiation carcinogenesis from galactic cosmic radiations (GCR). Our approach is to use the existing data on Harderian gland tumorigenesis from investigations of individual high-atomic number, and high-energy (HZE) ion beams to predict outcome of mixed ion chronic GCR exposures, and then measure Harderian gland and lung dose-dependent tumor data for such exposures. We will compare some of the measured cancer prevalence outcomes of a GCR-simulated exposure with the theoretical predictions based on experiments with individual ion beams. Our main hypothesis is that HZE carcinogenic effects are LET- (linear energy transfer) and dose-dependent and not additive at low particle doses due to the contribution of non-targeted effects. Completion of the proposed GCR studies using a mouse strain that has already been well characterized with a low spontaneous tumor background will allow a significant test of modeling approaches used to estimate carcinogenesis risk from a simulated GCR beamline.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Completion of our research on predicting and measuring HZE tumorigenic effects to test the hypothesis that HZE carcinogenic effects and LET- and dose-dependent are not additive at low particle doses due to the contribution of non-targeted effects will allow a significant test of modeling approaches used to estimate carcinogenic risk from single-ion and mixed radiation fields. This work will also benefit estimation of risk of tumorigenesis from mixed radiation fields on Earth.
Task Progress:	Note: The Principal Investigator (PI) has retired. See Bibliography for resulting publications since last report (Ed., 8/4/25).
Bibliography Type:	Description: (Last Updated: 02/12/2026)
Articles in Peer-reviewed Journals	Mairani A, Mein S, Blakely EA, Debus J, Durante M, Ferrari A, Fuchs H, Georg D, Grosshans DR, Guan F, Haberer T, Harrabi SB, Horst F, Inaniwa T, Karger CP, Mohan R, Paganetti H, Parodi K, Sala PR, Schuy C, Tessonnier T, Titt U, Weber U. "Roadmap: Helium ion therapy." Phys Med Biol. 2022 Aug 5;67(15). http://dx.doi.org/10.1088/1361-6560/ac65d3 ; PMID: 35395649 , Aug-2022
Articles in Peer-reviewed Journals	Chang PY, Bakke J, Rosen CJ, Bjornstad KA, Mao JH, Blakely EA. "Heavy-ion-induced lung tumors: Dose- & LET-dependence." Life (Basel). 2022 Jun 17;12(6):907. https://doi.org/10.3390/life12060907 ; PMID: 35743938; PMCID: PMC9225356 , Jun-2022
Articles in Peer-reviewed Journals	Huff JL, Poignant F, Rahmanian S, Khan N, Blakely EA, Britten RA, Chang P, Fornace AJ, Hada M, Kronenberg A, Norman RB, Patel ZS, Shay JW, Weil MM, Simonsen LC, Slaba TC. "Galactic cosmic ray simulation at the NASA Space Radiation Laboratory–Progress, challenges and recommendations on mixed-field effects." Life Sci Space Res. 2023 Feb;36:90-104. https://doi.org/10.1016/j.lssr.2022.09.001 ; PMID: 36682835 , Feb-2023
Articles in Peer-reviewed Journals	Leon Parada KN, Lawson G, Chang P, Blakely EA, Bajwa L, Gramajo-Aponte K, Getze S, Luderer U. "Irradiation with a mixed heavy ion beam induces ovarian follicle loss and dose-dependent mixed ovarian tumor development." Biol Reprod. 2026 Jan 21:ioag019. https://doi.org/10.1093/biolre/ioag019 ; PMID: 41562572 , Jan-2026